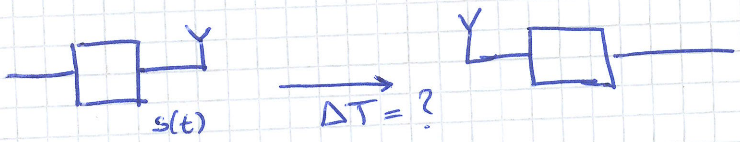
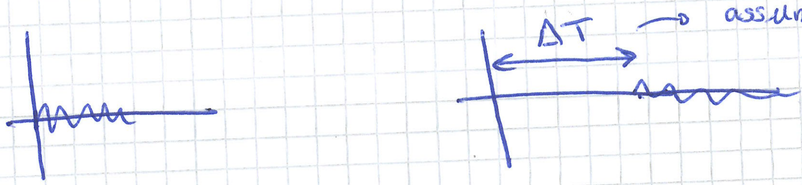
①

20/09/2024

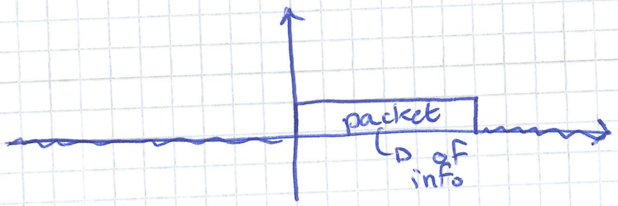
Wireless Receivers: Frame Synchronization





→ assuming both have same time ref

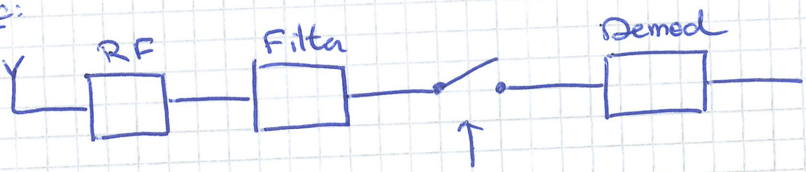
("–" delay could be possible)



is there packet or not ?

↳ simplifies synch

<u>Sync:</u>



We need to decide: When should I sample ?

<u>Frame synch</u>

T_x : $\underbrace{01001}_{\text{preamble}}$ x x x

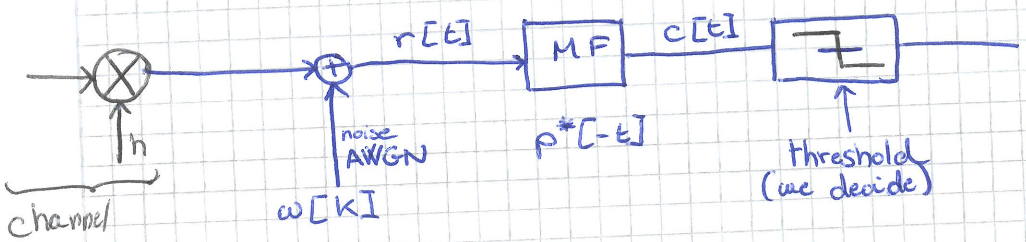
↳ uses same mod scheme

R_x : ~ ~ ~ ~ ~ $\underline{01001}$

preamble detected !

MF : matching Filter

<u>AWGN</u>



h = unknown

low h → low threshold & vice-versa → Threshold = f (h)

→ this is problematic, we want to find a general value indep of "h".

$\tilde{p}[i] = p[N_p - 1 - i]$ — N_p: preamble length

$$\underbrace{c[t] = \sum_{i=0}^{N_p-1} \tilde{p}[i]\, r[t-i]}_{\text{Correlation}}$$

Match Filter & Correlation are the same

Preamble DESIGN

No noise

BPSK $\{\pm 1\}$ $p[t]$

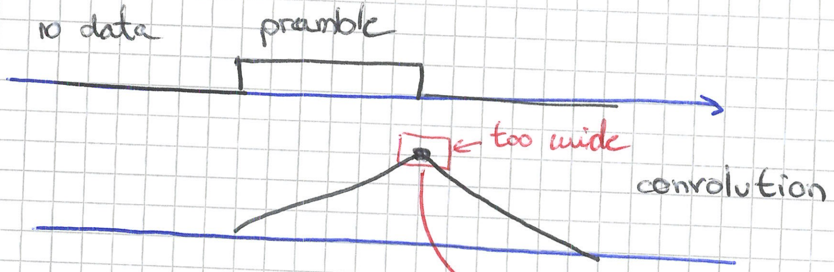
0 $d[t]$ (we are ignoring data now)

$$r[t] = \begin{cases} d[t] & t < t_0 \\ p[t - t_0] & t_0 < t < t_0 + N_p - 1 \\ d[0] & t_0 > N_p - 1 \end{cases}$$

① Const preamble $(p[t] + w[t]) * \underbrace{p^*[t]}_{\text{we agreed on } p[t]}$

(Correlation & Convolution are the same)

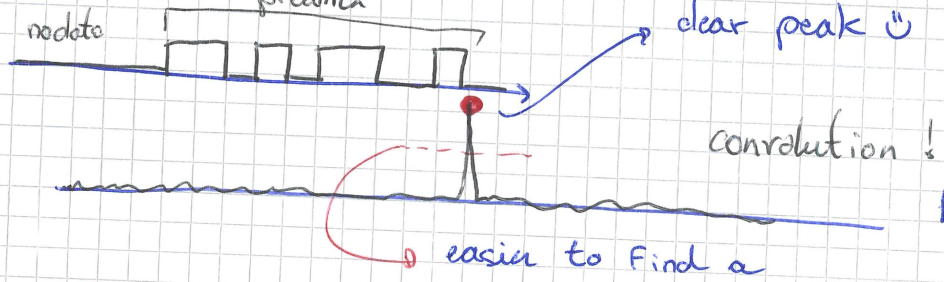
IMPORTANT



we want this point as it will indicate the start of the signal

⚠ here the problem is that @ max point we have many points close together! → recall in reality there is noise

What WE WANT!



we want a very precise AUTO-CORR Function in practice we will use p^* in the convolution to make the Im part of the multiplication "+". The objective is to mutiply the signal we receive w/ the preamble and check if the integral of what we get is OVER the threshold.

Synchronization

Demod : $p(a = x \mid r)$

$\hookrightarrow p(r \mid a = x)$ Hypothesis test

Frame sync : $\rightarrow$ 2 Hypothesis (or 2 possible scenarios)

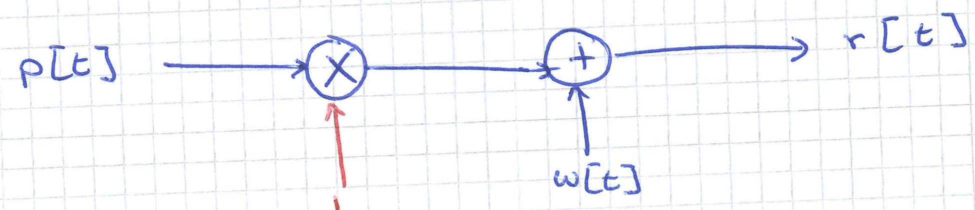
$\mathcal{H}_0$: no frame / this is wrong time $\rightarrow$ no detected preamble

$\mathcal{H}_1$: Sync $\rightarrow$ detected preamble

R_x signal:

$$\underline{r}[t] = \left[r[t], r[t+1], \dots, r[t+N_p-1] \right]$$

so we use a more $\mathbb{C}$ channel :



(channel modelled as scalar)

$\mathcal{H}_0$: no preamble $\underline{r_0}[t] = \underline{w}[t]$

$\mathcal{H}_1$: w/ preamble $\underline{r_1}[t] = h \cdot \underline{p}[t] + \underline{w}[t]$

I want to compare $\underline{r_0}$ and r_1

$$L_G(\underline{r}) = \frac{p(\underline{r} \mid \mathcal{H}_1)}{p(\underline{r} \mid \mathcal{H}_0)} > \gamma$$

Generalized Likelihood estimator

$$p(\underline{r} \mid \mathcal{H}_0) = \frac{1}{(\pi^2 \sigma_0^2)^{N_p}} e^{-\frac{\|\underline{r}\|^2}{\sigma_0^2}}$$

$\|\underline{r}\|^2 \rightarrow r_1^2 + r_2^2 + \dots +$

$N_p \rightarrow$ as we have N_p samples $\square^{N_p}$

σ_0 is the same, What is its value ?

Estimation of σ_0^2

$$\sigma_0^2 = \|r\|^2 \frac{1}{N_p}$$

Distribution of $\underline{r}$ assuming $\mathcal{H}_1$

For 1 sample $h\, p_\alpha + w_\alpha \sim \frac{1}{(\pi \sigma_1)^2} e^{-\frac{|r_\alpha - h p_\alpha|^2}{\sigma_1^2}}$

$$p(\underline{r} \mid \mathcal{H}_1) = \frac{1}{(\pi\sigma_1)^{2N_p}} e^{-\frac{\|\underline{r} - h\,\underline{p}\|^2}{\sigma_1^2}}$$

$\rightarrow$ we still don't know σ_1 (we will try to estimate it)

as ~~a~~ signal is not zero mean $\rightarrow$ $\sigma_1^2 = \frac{1}{N_p}\|\underline{r}\|^2 - \frac{\|h\,\underline{p}\|^2}{N_p}$

$$\frac{\|h\,\underline{p}\|^2}{N_p} = \frac{|h|^2 \|\underline{p}\|^2}{N_p}$$

$\rightarrow$ only values $\pm 1 \Rightarrow \|\underline{p}\|^2 = N_p$

$\rightarrow = |h|^2$

$$\sigma_1^2 = \frac{1}{N_p}\|\underline{r}\|^2 - |h|^2$$

estimate h from r

$\hat{h} = \underset{h}{\operatorname{argmax}}\left(p(\underline{r} \mid \mathcal{H}_1, h)\right)$ maximized when $\min\left(\|r - h \cdot \underline{p}\|^2\right)$

$$\rightarrow \hat{h} = \underset{h}{\operatorname{argmin}}\left(\underbrace{\|\underline{r} - h \cdot \underline{p}\|^2}_{z}\right)$$

develop the square

$$z = \|\underline{r} - h\,\underline{p}\|^2 = \underbrace{\|\cancel{r}\|^2}_{\text{indep } h} - \underline{r}^H \underline{p}\, h - (h\,\underline{p})^H \underline{r} + |h|^2 \underbrace{\|\underline{p}\|^2}_{N_p}$$

Find "h" that minimizes "z"

$$h = |h|\, e^{j\phi_h}$$

3

$$-\underline{r}^H \underline{p} h - h \underline{p}^H \underline{r} = -2 \cdot \mathrm{Re}\{ \underbrace{\underline{p}^H \underline{r} |h| e^{-j\phi_n}} \}$$

we want to maximize this to make the other as small as possible

$$\hat{=} |h|^2 N_p - 2|h| \underline{p}^H \underline{r} \quad \Big| \frac{\partial (\dots)}{\partial h} = 0 \text{ to Find "h"}$$

$$\Rightarrow |\hat{h}| = \frac{|p^H r|}{N_p} \rightarrow \hat{h} = \frac{\underline{p}^H \underline{r}}{N_p}$$

here we can also ~~sync~~ estimate the channel h.

now, we can go back to Finding $\sigma_1^2 = \frac{1}{N_p} \|r\|^2 - |h|^2$

using our estimate, we find

$$\sigma_1^2 = \frac{1}{N_p}\left(\|r\|^2 - \frac{|p^H r|^2}{N_p} \right)$$

Generalized Likelihood estimate

$$L(\underline{r}) = \frac{p(\underline{r} | \mathcal{H}_1)}{p(\underline{r} | \mathcal{H}_0)} = \left(\frac{\frac{\pi}{N_p} |r|^2}{\frac{\pi}{N_p}\left(\|r\|^2 - \frac{|p^H r|^2}{N_p} \right)} \right)^{N_p}$$

$$= \left(\frac{1}{1 - \underbrace{\frac{|\underline{p}^H \underline{r}|^2}{|r|^2 N_p}}_{\text{this is the only relevant part}}} \right)^{N_p} > \gamma$$

we adjust (use $\neq \gamma$) to eventually get

$$\frac{|\underline{p}^H \underline{r}|^2}{|\underline{r}|^2} > \gamma'' \quad \Bigg\} \text{ Scaled MF output}$$

due to h

Squared mag of the MF output